Spacetime from Self-Modeling: Einstein's Equations from Entanglement Equilibrium on a Self-Modeling Lattice

Bryan Ehrlich[1]

[1] *Independent researcher*

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We show that the locality of self-modeling—the principle that a physical system contains a faithful model of itself, updated through its boundary—yields Einstein's field equations as the leading-order effective description at long wavelengths, given lattice topology and antiferromagnetic coupling as inputs. Self-modeling forces $M_n(\mathbb{C})^{sa}$ quantum mechanics [1]. The self-modeling interaction Hamiltonian $H = \sum_{\langle x,y \rangle} J\, F_{xy}$, forced by diagonal $U(n)$ covariance, produces area-law entanglement entropy. We present two routes to

$$G_{ab} + \Lambda\, g_{ab} = 8\pi G\, T_{ab}$$

with $G = 1/(4\eta)$: (A) Jacobson's entanglement equilibrium argument [2], exact in the conformal regime; and (B) Lovelock's uniqueness theorem [3], which forces Einstein's equation in $d \geq 2$ spatial dimensions without requiring conformal symmetry. The maximal vacuum entanglement hypothesis (MVEH), traditionally an additional assumption, is identified as the only available vacuum selection criterion in a pre-geometric framework via the Connes–Rovelli thermal time hypothesis [4]. We present numerical verification on small lattices ($N = 8$–20) and precisely identify remaining inputs: the lattice topology, the antiferromagnetic coupling sign, and the Wilsonian continuum limit.

I. INTRODUCTION

Self-modeling—the condition that a physical system contains a faithful model of itself, updated through its boundary—forces the algebraic structure of quantum mechanics [1]. We show that it also forces general relativity. Specifically, Einstein's field equations emerge as the leading-order infrared effective description of a lattice of self-modeling systems in the Wilsonian continuum limit.

The idea that spacetime geometry is encoded in quantum entanglement has a rich history. Van Raamsdonk argued that entanglement is the glue that holds spacetime together [5]. Jacobson showed that the Einstein equation can be derived as an equation of state from thermodynamic considerations at local causal horizons [6], and later sharpened this to an entanglement equilibrium argument requiring only area-law entanglement entropy, the entanglement first law, and the maximal vacuum entanglement hypothesis (MVEH) [2]. Lashkari, McDermott, and Van Raamsdonk demonstrated that the entanglement first law in holographic conformal field theories implies the linearized Einstein equation [7], and Faulkner *et al.* extended this to the full nonlinear equations in the holographic setting [8]. Cao, Carroll, and Michalakis showed that spatial geometry can be recovered from the entanglement structure of a finite-dimensional Hilbert space [9].

These results leave open a central question: *what ultraviolet completion produces the entanglement structure that Jacobson's argument requires?* The holographic results of [7, 8] assume AdS/CFT; the Jacobson argument [2] is UV-agnostic. We provide a specific, non-holographic UV completion: a lattice of self-modeling quantum systems whose interaction Hamiltonian is forced by the algebraic structure of quantum theory itself.

The main result can be stated as follows. Self-modeling is the sole physical premise. The lattice topology $G = (V, E)$ and the sign of the coupling ($J > 0$, antiferromagnetic) are inputs. The Wilsonian continuum limit is standard methodology. We present two routes to Einstein's equations: an exact derivation via Jacobson's entanglement equilibrium in the conformal regime, and a uniqueness argument via Lovelock's theorem that extends to all $d \geq 2$ under the assumption that entanglement perturbations yield a tensorial geometric equation.

The maximal vacuum entanglement hypothesis—that the vacuum maximizes entanglement entropy at fixed stress-energy expectation value—traditionally appears as an additional assumption in Jacobson's framework. We argue that in the present context it is a *structural identification*: in the absence of pre-existing geometry, the modular flow of the self-modeling lattice state defines the emergent geometric flow, following the thermal time hypothesis of Connes and Rovelli [4]. The state whose modular flow defines a smooth emergent geometry necessarily satisfies MVEH—the "vacuum" is the geometry-defining state. This is the only available identification in a pre-geometric framework, though we do not prove existence or uniqueness of such a state. We note that Sorce [10] has shown that geometric modular flow requires conformal symmetry; this caveat is addressed in Sec. IV, where we observe that the $SU(n)$ covariance of the self-modeling lattice places its infrared description in the class of conformal field theories for which the identification is clean.

This paper does not derive Newton's constant G (which depends on the lattice entropy density η via $G = 1/(4\eta)$), the spacetime dimension (which is set by the lattice topology), or the cosmological constant Λ (which

appears as an undetermined integration constant). The sign chain producing attractive gravity assumes the null energy condition (NEC).

The complete derivation chain, from self-modeling to Einstein's equations, is summarized in Table I.

The novel contribution of this work is the assembly of Links L1–L5 as a UV completion, with L2–L4 being new results: the forced Hamiltonian (L2), emergent causal structure (L3), and area-law bounds (L4). L1 is established in Paper 5 [1] and L5 is a standard identity of quantum information theory. We present two routes from L1–L7 to Einstein's equations: Route A applies Jacobson's entanglement equilibrium argument [2] (exact in the conformal regime), and Route B uses Lovelock's uniqueness theorem [3] (valid in all $d \geq 2$ spatial dimensions without requiring conformal symmetry). L7 is reframed from an assumption to a structural identification.

The paper is organized as follows. Section II defines the self-modeling lattice: the local algebra $M_n(\mathbb{C})^{sa}$ at each site, the SWAP interaction Hamiltonian forced by diagonal $U(n)$ covariance, and the emergent causal structure from the Lieb-Robinson velocity [11, 12]. Section III establishes sub-volume entanglement scaling: the WVCH thermal mutual information bound [13], known ground-state entanglement properties of the Heisenberg model [14, 15], and the entanglement first law combined with modular Hamiltonian locality. Section IV presents the thermal time hypothesis and its application to the self-modeling lattice, recasting the MVEH assumption as a structural identification. Section V derives Einstein's equations via two routes: Jacobson's entanglement equilibrium argument [2] in the conformal regime (Route A), and Lovelock's uniqueness theorem [3] in $d \geq 2$ (Route B). Section VI presents numerical verification on small lattices ($N = 8$–20): exact diagonalization benchmarks, area-law scaling, modular Hamiltonian locality, and a qualitative MVEH check. Section VII identifies remaining gaps, compares with related work, and outlines future directions.

II. THE SELF-MODELING LATTICE

A. Self-modeling forces quantum mechanics

Paper 5 [1] established that faithful self-modeling—the condition that a physical system B contains a subsystem M that tracks the state of B through their shared boundary—forces the local state space to be $M_n(\mathbb{C})^{sa}$, the self-adjoint part of the $n \times n$ complex matrix algebra, equipped with the Lüders sequential product

$$a \,\&\, b = a^{1/2}\, b\, a^{1/2}. \tag{1}$$

The real and quaternionic alternatives are excluded by local tomography: only complex quantum mechanics has $\dim(V_{BM}) = \dim(V_B) \cdot \dim(V_M)$. The sequential product is the unique product compatible with axioms S1–S7 of Ref. [16], and it inherits $U(n)$ covariance from the self-modeling structure. We do not re-derive these results here; they serve as the starting point for the present paper.

B. Lattice definition

We place self-modeling systems on a lattice described by an undirected graph $G = (V, E)$, where V is a countable set of sites and $E \subseteq \binom{V}{2}$ is the set of edges encoding nearest-neighbor adjacency. At each site $x \in V$ the local algebra is

$$A_x = M_n(\mathbb{C}), \tag{2}$$

and for a finite region $\Lambda \subset V$ the regional algebra is

$$A_\Lambda = \bigotimes_{x \in \Lambda} A_x \cong M_{n^{|\Lambda|}}(\mathbb{C}). \tag{3}$$

The quasi-local algebra $\mathfrak{A} = \overline{\bigcup_{|\Lambda|<\infty} A_\Lambda}^{\|\cdot\|}$ is a uniformly hyperfinite (UHF) C*-algebra of type n^∞.

Background dependence.—The graph topology $G = (V, E)$ is an irreducible *input* to this construction. Self-modeling determines the metric (encoded in the coupling strengths and the resulting entanglement structure) on a given topology, but not the topology itself. This is analogous to general relativity, where Einstein's equations determine the metric on a given differentiable manifold without determining its topology. The spatial dimension d is set by the graph structure: a d-dimensional regular lattice yields d emergent spatial dimensions.

C. Hamiltonian construction

The interaction between adjacent sites is encoded in a two-site Hamiltonian $h_{xy} \in A^{sa}_{\{x,y\}}$ for each edge $\{x, y\} \in E$. Self-modeling locality—the constraint that model and body interact only through their shared boundary—implies that $h_{xy} = 0$ for non-adjacent pairs ($\{x, y\} \notin E$), mapping self-modeling locality directly to nearest-neighbor Hamiltonian locality [1].

The Lüders sequential product (1) is covariant under the adjoint action of $U(n)$:

$$U(a \,\&\, c)U^\dagger = (UaU^\dagger) \,\&\, (UcU^\dagger) \qquad \forall\, U \in U(n). \tag{4}$$

The composite sequential product $(a \otimes b) \,\&\, (c \otimes d) = (a \,\&\, c) \otimes (b \,\&\, d)$ is covariant under $U(n) \times U(n)$ (independent rotations at each site). The coupling, however, physically connects the two sites through a shared boundary; basis independence requires that the same rotation be applied to both sites. This constrains h_{xy} to be invariant under the *diagonal* $U(n)$:

$$(U \otimes U)\, h_{xy}\, (U \otimes U)^\dagger = h_{xy} \qquad \forall\, U \in U(n). \tag{5}$$

TABLE I. Derivation chain from self-modeling to Einstein's equations. "Status" gives the logical standing of each link. "Novel?" indicates what is new in this work versus prior results. L7 is reframed as a structural identification via the Connes–Rovelli thermal time hypothesis [4], not an independent assumption.

Link	Statement	Status	Section	Novel?	Key input
L1	Self-modeling forces $M_n(\mathbb{C})^{sa}$ with Lüders product	Proven	—	Ref. [1]	None
L2	Lattice Hamiltonian $H = \sum J\, F_{xy}$ forced by $U(n)$ covariance	Derived	II	This work	Lattice topology,
L3	Lieb-Robinson velocity v_{LR} gives emergent causal structure	Derived	II	This work	L2
L4	Area-law entanglement entropy	Established	III	This work	L2, L3
L5	Entanglement first law: $\delta S = \delta\langle K_A\rangle$	Exact identity	III	Standard QI	None
L6	Wilsonian continuum limit	Standard methodology	V	Standard	L2
L7	Modular flow defines geometry (thermal time hypothesis)	Structural ident.	IV	Reframed	Connes–Rovelli [4
L8a	$G_{ab} + \Lambda\, g_{ab} = 8\pi G\, T_{ab}$ (Jacobson route)	Derived (conformal)	V A	Jacobson [2] applied	L1–L7, conformal
L8b	$G_{ab} + \Lambda\, g_{ab} = 8\pi G\, T_{ab}$ (Lovelock route)	Uniqueness ($d \geq 2$)	V B	This work	L1–L7, Lovelock

Full $U(n) \times U(n)$ invariance would kill all coupling, leaving $h_{xy} \propto \mathbb{1} \otimes \mathbb{1}$.

By **Schur–Weyl duality** for the symmetric group S_2 acting on $(\mathbb{C}^n)^{\otimes 2}$, this space decomposes under the diagonal $U(n)$ action into two irreducible representations: $\mathrm{Sym}^2(\mathbb{C}^n) \oplus \wedge^2(\mathbb{C}^n)$. By Schur's lemma the commutant of $\{U \otimes U\}$ is $\mathrm{span}\{\mathbb{1}, F\}$, where F is the SWAP operator defined by

$$F_{xy}|i\rangle|j\rangle = |j\rangle|i\rangle. \tag{6}$$

Therefore the most general diagonal-$U(n)$-invariant self-adjoint interaction is

$$h_{xy} = \alpha\, \mathbb{1} \otimes \mathbb{1} + J\, F_{xy}, \tag{7}$$

with $\alpha, J \in \mathbb{R}$. The identity term shifts the energy zero; setting $\alpha = 0$, the total lattice Hamiltonian is

$$\boxed{H = \sum_{\langle x,y\rangle \in E} J\, F_{xy}.} \tag{8}$$

The interaction is *forced*: the form $h_{xy} = J\, F_{xy}$ is the unique diagonal-$U(n)$-invariant coupling (up to the overall energy scale J). Any anisotropy would break the $U(n)$ covariance inherited from the Lüders product. For $n = 2$, the SWAP operator takes the familiar form $F_{xy} = \frac{1}{2}(\mathbb{1} \otimes \mathbb{1} + \vec{\sigma} \cdot \vec{\sigma})$, and the Hamiltonian (8) reduces to the isotropic Heisenberg model.

The sign of J is *not* determined by the self-modeling constraints: $J > 0$ gives the antiferromagnetic (AFM) Heisenberg model, $J < 0$ gives the ferromagnetic (FM) model. Both signs are compatible with the algebraic structure. However, the FM ground state ($J < 0$) is a product state with $S(A) = 0$ for all subsystem sizes, producing no entanglement structure and hence no emergent geometry. Only the AFM case ($J > 0$) supports the nontrivial entanglement that drives the Jacobson argument. We therefore work with $J > 0$ throughout; this is an input to the derivation, not forced by self-modeling.

D. Lieb–Robinson bounds and emergent causal structure

The nearest-neighbor Hamiltonian (8) with finite coupling $\|h_{xy}\| = |J|$ satisfies the hypotheses of the Lieb–Robinson theorem [11, 12]. For any local observables A_x and B_y supported at sites x and y separated by graph distance $d(x, y)$, the Heisenberg-picture commutator is bounded by

$$\|[\tau_t(A_x),\, B_y]\| \leq C_{\mathrm{LR}}\, \|A\|\, \|B\|\, e^{-\mu(d(x,y) - v_{\mathrm{LR}}|t|)}, \tag{9}$$

where $\tau_t(A) = e^{iHt} A\, e^{-iHt}$ is the Heisenberg evolution, $\mu > 0$ is a spatial decay constant, C_{LR} is a dimension-dependent prefactor, and v_{LR} is the **Lieb–Robinson velocity**—the maximum speed of information propagation on the lattice.

For the self-modeling Hamiltonian on a lattice with coordination number z, the Nachtergaele–Sims framework [12] with exponential decay function $F_a(r) = e^{-ar}$ at $a = 1$ gives

$$v_{\mathrm{LR}} = \frac{4ze|J|}{e - 1}. \tag{10}$$

For the one-dimensional chain ($z = 2$), this is $v_{\mathrm{LR}} = 8e|J|/(e - 1) \approx 12.66\, |J|$. Crucially, v_{LR} is *independent of the local dimension* n: the SWAP operator F satisfies $\|F\| = 1$ for all $n \geq 2$, so the Lieb–Robinson velocity depends only on the coupling strength $|J|$ and the lattice coordination number z.

The bound (9) defines an effective *light cone* on the lattice: correlations outside the cone ($d(x, y) > v_{\mathrm{LR}}|t|$) are exponentially suppressed. In the continuum limit (Sec. V), v_{LR} is identified with the emergent speed of light c.

Causal structure, not Lorentz invariance.—The Lieb–Robinson bound provides an emergent *causal structure*—a finite maximum signaling speed—but not Lorentz invariance. The lattice has a preferred frame (the rest frame of the lattice). Lorentz invariance of the low-energy effective theory is an output of the Wilsonian

continuum limit (Sec. IV D), not a property of the lattice itself. The Lieb–Robinson velocity $v_{\rm LR}$ sets the *scale* of the emergent speed of light; the *symmetry* (Lorentz covariance) is a property of the long-wavelength effective theory, realized when lattice artifacts are suppressed at $L \gg a$.

III. AREA-LAW ENTANGLEMENT

The Jacobson entanglement equilibrium argument [2] requires two inputs from the UV theory: (J1) the entanglement entropy variation δS scales with the boundary area, not the volume; and (J2) the entanglement first law $\delta S = \delta\langle K_A\rangle$ holds exactly. We now establish both for the self-modeling lattice.

A. Entanglement first law

For any quantum state ρ with reduced state $\rho_A = \mathrm{Tr}_B(\rho)$ and modular Hamiltonian $K_A = -\ln\rho_A$, the first-order change in the von Neumann entropy under a perturbation $\rho \to \rho + \delta\rho$ with $\mathrm{Tr}(\delta\rho) = 0$ is

$$\boxed{\delta S(A) = \delta\langle K_A\rangle.} \tag{11}$$

This is an *exact identity* of quantum information theory [7, 17]: expanding $S = -\mathrm{Tr}(\rho_A \ln\rho_A)$ to first order in $\delta\rho_A$ and using $\mathrm{Tr}(\delta\rho_A) = 0$ gives $\delta S = -\mathrm{Tr}(\delta\rho_A \ln\rho_A) = \mathrm{Tr}(\delta\rho_A\, K_A) = \delta\langle K_A\rangle$. No approximations or additional assumptions are involved. This establishes Jacobson input (J2).

B. Area-law entanglement bounds

Consider a bipartition of the lattice $V = A \cup B$ with boundary $\partial A = \{\langle x, y\rangle \in E : x \in A,\, y \in B\}$ of size $|\partial A|$. We establish sub-volume entanglement scaling from complementary perspectives.

1. Thermal mutual information (WVCH)

Assumption: The state is a Gibbs state $\rho_\beta = e^{-\beta H}/Z$ at inverse temperature $\beta = 1/T > 0$.

For any Gibbs state of a local Hamiltonian, the quantum mutual information $I(A{:}B) = S(A) + S(B) - S(AB)$ satisfies the Wolf–Verstraete–Cirac–Hastings (WVCH) bound [13]:

$$I(A{:}B) \le 2\beta \sum_{\substack{X:\, X\cap A\neq\emptyset \\ X\cap B\neq\emptyset}} \|\Phi(X)\|. \tag{12}$$

For the self-modeling Hamiltonian (8) with nearest-neighbor coupling $\|h_{xy}\| = |J|$, this reduces to

$$\boxed{I(A{:}B) \le 2\beta\, |\partial A|\, |J|.} \tag{13}$$

The mutual information scales with the boundary area $|\partial A|$, not the volume $|A|$. The bound holds in all spatial dimensions, requires no spectral gap, and depends on $|J|$ (not $\mathrm{sign}(J)$), so it applies equally to the AFM and FM cases.

2. Ground-state entanglement of the Heisenberg model

We are not proving a new area-law theorem. We observe that the Hamiltonian self-modeling forces—the isotropic Heisenberg model—has well-studied entanglement properties (see [18] for a comprehensive review).

One dimension, gapless ($n = 2$, AFM): The spin-$\frac{1}{2}$ Heisenberg antiferromagnet flows to the $SU(2)_1$ WZW CFT with central charge $c = 1$. The Calabrese–Cardy formula [14] gives $S(L) = \frac{c}{3}\ln(L) + \mathrm{const}$, a logarithmic correction to the strict area law. This is sub-volume: S grows as $\ln L$, not L. For the Jacobson argument, sub-volume scaling suffices (Sec. III B 3).

One dimension, gapped ($n \ge 3$, AFM): For integer spin ($n \ge 3$), the Haldane conjecture [19] (proved for large n [20]) gives a spectral gap. Hastings' area-law theorem [15] then guarantees $S(A) \le$ const for one-dimensional gapped ground states, a strict area law.

Two and higher dimensions: The $d \ge 2$ AFM Heisenberg model is expected to have Néel order (spectral gap above the symmetry-broken ground state) [18]. Area-law entanglement for gapped ground states in $d \ge 2$ is supported by extensive numerical evidence and physical arguments, but a rigorous proof for all local Hamiltonians in $d \ge 2$ remains open [18]. Our numerical results (Sec. VI) on the 4×4 lattice are consistent with area-law scaling ($R^2 = 0.885$ for boundary, 0.491 for volume).

The maximum entanglement that a single boundary bond ($\mathbb{C}^n \otimes \mathbb{C}^n$) can carry is

$$S_{\rm bond} \le \log n, \tag{14}$$

by the Schmidt decomposition. This per-bond capacity sets the entanglement density scale $\eta_{\rm lattice} \le \log n$ used in the parameter identification of Sec. V A 5. Note that $S(A)$ for the full ground state is *not* bounded by $\log(n)\cdot|\partial A|$: the gapless AFM chain has $S(A) \sim (c/3)\ln L$, which grows without bound as $L \to \infty$ while $|\partial A|$ remains fixed. The Jacobson argument requires sub-volume scaling of δS, not a strict area-law bound on $S(A)$ itself (Sec. III B 3).

3. Perturbative δS (modular Hamiltonian locality)

The Jacobson argument requires not that $S(A)$ itself obeys an area law, but that the *variation* δS under small perturbations scales with $|\partial A|$. From the entanglement first law (11),

$$\delta S = \delta\langle K_A\rangle. \tag{15}$$

If the modular Hamiltonian $K_A = -\ln \rho_A$ is concentrated near the boundary ∂A (i.e., its matrix elements decay rapidly with distance from the boundary), then $\delta\langle K_A\rangle \sim O(|\partial A|)$ for local perturbations, giving

$$\delta S \sim O(|\partial A|). \tag{16}$$

Modular Hamiltonian locality is physically motivated by the Bisognano–Wichmann theorem [21] (the modular Hamiltonian of the vacuum in a Rindler wedge is the boost generator, localized at the entangling surface) and by Peschel's result for free lattice fermions [22] (K_A is a nearest-neighbor hopping operator on the entangling surface). For the self-modeling Hamiltonian, numerical evidence (Sec. VI) supports this: the short-range fraction of K_A is 0.9993 for the antiferromagnetic Heisenberg chain at $N = 16$.

This perspective is the weakest of the three, since modular Hamiltonian locality is not proven in general for the self-modeling lattice. However, it is the most directly relevant for the Jacobson argument: it establishes that δS is an area-law quantity even when $S(A)$ itself has logarithmic corrections (as in the gapless AFM case).

C. What this establishes

The perspectives complement each other:

- **Thermal states:** $I(A{:}B) \leq 2\beta|J|\,|\partial A|$ (WVCH, no gap required, strict area law on mutual information).
- **Ground states:** Sub-volume scaling ($S \sim \ln L$ in gapless 1D, strict area law in gapped phases). Not a strict area law in the gapless case, but sufficient for Jacobson.
- **Perturbatively:** $\delta S \sim O(|\partial A|)$ (modular Hamiltonian locality, numerically verified).

The third perspective is the most directly relevant for the Jacobson argument, which requires area-law scaling of δS, not of S itself. Combined with the entanglement first law (11), which establishes Jacobson input (J2), the self-modeling lattice provides the two UV inputs required by Jacobson's entanglement equilibrium argument.

Note on genericity.—A generic pure state on a local lattice has *volume-law* entanglement; area-law scaling is a special property of ground states (and low-energy states) of local Hamiltonians. The self-modeling lattice produces area-law-like entanglement not because of a generic locality argument, but because the Hamiltonian it forces (the Heisenberg model) has well-characterized ground-state entanglement in the condensed matter literature [18].

IV. ENTANGLEMENT EQUILIBRIUM AND THE THERMAL TIME HYPOTHESIS

The Jacobson entanglement equilibrium argument [2] requires a third input beyond area-law entanglement and the entanglement first law: the maximal vacuum entanglement hypothesis (MVEH), the condition that the vacuum maximizes entanglement entropy at fixed stress-energy expectation value. In the original Jacobson framework, MVEH is an additional assumption. We argue that in the present context—where there is no pre-existing geometry—it is a *structural identification*: the only available vacuum selection criterion, rather than an independent physical assumption.

A. Modular flow and the thermal time hypothesis

The Tomita–Takesaki theorem guarantees that every faithful normal state ω on a von Neumann algebra $\mathcal{M}$ defines a canonical one-parameter automorphism group $\sigma_t^\omega : \mathcal{M} \to \mathcal{M}$, the *modular automorphism group* (or modular flow) [23]. The flow is determined entirely by the pair $(\mathcal{M}, \omega)$: no external Hamiltonian or time parameter is needed.

Connes and Rovelli [4] proposed the *thermal time hypothesis*: in a generally covariant quantum theory—one with no preferred notion of time—the physical time evolution *is* the modular flow. Time is not a background parameter; it is selected by the state. This proposal was motivated by the observation that the modular flow of a thermal state ω_β at inverse temperature β is precisely the physical time evolution rescaled by β: the thermal state "knows" its own dynamics.

For the self-modeling lattice, there is no pre-existing spacetime geometry. There is no background metric, no pre-assigned light cone, no external time parameter. The lattice carries an algebra $\mathfrak{A}$ (the quasi-local algebra of Sec. II B) and a state. The modular flow of this state is the *only* candidate for a local time evolution in the absence of external structure.

Near the boundary ∂A of a subregion A, the modular flow plays the role of a local boost generator. In quantum field theory on a fixed spacetime, this is the content of the Bisognano–Wichmann theorem [21]: the modular Hamiltonian of the Minkowski vacuum restricted to a Rindler wedge is the Lorentz boost generator, and the modular flow is the boost. Here, we *define* the emergent geometric structure from the modular flow, rather than deriving the modular flow from a pre-existing geometry.

B. MVEH as structural identification

The maximal vacuum entanglement hypothesis states that the vacuum maximizes $S(B)$ for any small geodesic ball B at fixed stress-energy expectation value $\langle T_{ab}\rangle$.

When a background spacetime exists, this is a substantive physical assumption: one must check that the vacuum of a given quantum field theory actually satisfies this extremal condition.

In the self-modeling framework, there is no pre-existing geometry against which to test the hypothesis. The logical structure is different:

> In the absence of pre-existing geometry, the modular flow of the self-modeling lattice state defines the emergent geometric flow (thermal time hypothesis, Connes–Rovelli [4]). The state whose modular flow defines a smooth emergent geometry necessarily satisfies MVEH—the "vacuum" *is* the geometry-defining state.

We regard this as a *structural identification*: in a pre-geometric framework, there is no other candidate for the vacuum state, and the Connes–Rovelli thermal time hypothesis provides the only available identification of geometry with state.

This identification is strengthened by the entanglement–geometry connection established by Van Raamsdonk [5]: reducing entanglement between two regions increases their effective geometric distance; in the limit of zero entanglement, the geometry disconnects entirely. The converse—that *maximal* entanglement at fixed energy produces the smoothest, most connected geometry—implies that the geometry-defining state (the "vacuum") is precisely the state that maximizes entanglement at fixed stress-energy. This *is* the MVEH. The argument runs:

1. In a pre-geometric framework, "vacuum" means the state whose modular flow defines a smooth emergent geometry (Connes–Rovelli).

2. A smooth geometry requires maximal correlations across all regions (Van Raamsdonk: less entanglement $\to$ less connected geometry).

3. The state with maximal correlations at fixed $\langle T_{ab}\rangle$ is the maximum-entropy state—i.e., the state satisfying MVEH.

MVEH thus follows from combining the thermal time identification (Connes–Rovelli) with the entanglement–geometry correspondence (Van Raamsdonk). We do not claim this as a mathematical proof: step 2 relies on the physical intuition that entanglement is the "glue" of geometry, and we do not prove existence or uniqueness of the geometry-defining state. But the argument is stronger than a bare structural identification—it provides a physical mechanism connecting vacuum selection to entanglement maximization.

MVEH does not appear as a numbered assumption in this paper; it is the structural identification that selects which state defines the geometry.

C. The Sorce caveat

Sorce [10] proved that geometric modular flow—modular flow that acts as a local geometric symmetry (a conformal Killing flow)—requires conformal symmetry of the underlying quantum field theory. Non-isometric conformal Killing flows arise only in conformal field theories. This places a constraint on the thermal time identification: not every modular flow corresponds to a smooth geometric flow.

For the self-modeling lattice, this caveat is addressed by the $SU(n)$ symmetry of the Hamiltonian (8). The diagonal $U(n)$ covariance, combined with the nearest-neighbor structure, places the infrared description of the lattice in the class of $SU(n)$ symmetric spin models. For $n = 2$ with $J > 0$ (antiferromagnetic), the infrared theory is the $SU(2)_1$ Wess–Zumino–Witten conformal field theory with central charge $c = 1$ [24, 25]. For general n with $J > 0$, the $SU(n)$ Heisenberg model flows to the $SU(n)_1$ WZW CFT [26]. In these conformal theories, the modular Hamiltonian of the vacuum restricted to an interval takes the Casini–Huerta–Myers form [27], and the Sorce condition is satisfied.

We state honestly that Sorce's result provides *necessary* conditions for geometric modular flow; *sufficient* conditions remain an open question. The resolution via $SU(n)$ conformal symmetry in the infrared is clean for the one-dimensional case ($d = 1$), where the WZW CFT description is well established.

For $d \geq 2$, this resolution faces a genuine open problem. The $SU(n)$ Heisenberg antiferromagnet on the square lattice ($d = 2$) is expected to exhibit Néel order—spontaneous breaking of $SU(n)$ symmetry—rather than conformal behavior. The infrared theory is then a nonlinear sigma model with a mass gap (spin-wave gap), not a CFT. The conformal modular Hamiltonian (Sec. V A 3) would apply only in the window $a \ll \ell \ll 1/m$ (where m is the Néel gap scale), if such a window exists. This is the principal limitation of the present approach in dimensions $d \geq 2$: the paper's results are strongest in $d = 1$, where the WZW conformal structure is exact.

D. Wilsonian continuum limit

At length scales $L \gg a$ (lattice spacing), we assume that the lattice details become irrelevant by a Wilsonian universality argument: the long-wavelength physics is governed by the symmetry-allowed effective field theory, and the smooth Riemannian manifold (M, g_{ab}) emerges at long wavelengths.

The Wilsonian continuum limit is standard *methodology*, employed in lattice QCD, causal dynamical triangulations, and Regge calculus. However, its *applicability* to a given lattice model is not automatic. Lattice QCD benefits from asymptotic freedom, which guarantees a UV fixed point and controlled continuum limit. The self-modeling Hamiltonian has no such guarantee in

$d \geq 2$: whether the Heisenberg model on a d-dimensional lattice admits a continuum limit with the correct universality class is an open question, particularly for $d \geq 3$ where the model may be trivial or confining. A rigorous constructive proof is a hard open problem shared by all lattice quantum gravity programs. We frame the continuum limit as a physical argument, not a rigorous construction.

The Wilsonian picture suggests a natural perspective from the self-modeler: the compressed model M is finite-dimensional. Below the model's resolution (the lattice scale a), there is nothing to see. Smoothness is not a property of the lattice; it is a property of the observer's coarse-grained description at scales $L \gg a$. The continuum limit is the self-modeler's view.

In the continuum limit, the lattice parameters acquire physical identifications:

- The lattice spacing a maps to the Planck scale: $a \lesssim 2\ell_P\sqrt{\log n}$ (via the per-bond entropy bound (14) and $G = 1/(4\eta)$).
- The Lieb–Robinson velocity $v_{\rm LR}$ maps to the speed of light c.
- The lattice entropy density $\eta_{\rm lattice} \leq \log n$ per boundary bond maps to the continuum entropy density $\eta = \eta_{\rm lattice}/a^{d-1}$.

V. EINSTEIN'S EQUATIONS

We present two independent routes from the self-modeling lattice to Einstein's field equations, with complementary strengths. Route A (Sec. V A) applies Jacobson's (2016) entanglement equilibrium argument [2] using the conformal modular Hamiltonian; it is exact in the conformal regime ($d = 1$ WZW, or the window $a \ll \ell \ll 1/m$ in higher d). Route B (Sec. V B) uses Lovelock's uniqueness theorem [3, 28] to show that *any* tensorial equation relating entanglement perturbations to geometry perturbations must be Einstein's equation in $d \geq 2$ spatial dimensions. Route B does not require a conformal modular Hamiltonian.

A. Route A: Jacobson entanglement equilibrium (conformal regime)

1. Entropy decomposition

Consider a point p in the emergent spacetime manifold (M, g_{ab}) obtained from the Wilsonian continuum limit (Sec. IV D). Choose Riemann normal coordinates at p so that $g_{ab}(p) = \eta_{ab}$ and the Christoffel symbols vanish. Let B be a geodesic ball of proper radius ℓ centered at p in the spacelike hypersurface $\Sigma = \{x^0 = 0\}$, with $a \ll \ell \ll L_{\rm curv}$ (the curvature radius). The causal diamond $\mathcal{D}(B)$ is the domain of dependence of B.

The entanglement entropy of ball B decomposes as [2]

$$S = S_{\rm UV} + S_{\rm mat}, \tag{17}$$

where $S_{\rm UV} = \eta\,\mathcal{A}$ is the area-proportional UV contribution (with $\eta = 1/(4G)$ the entropy density per unit area and $\mathcal{A}$ the area of ∂B), and $S_{\rm mat}$ is the finite, state-dependent matter contribution. The UV entropy density η is a constant of the lattice structure; it does not depend on the state.

Under a first-order perturbation away from the maximally symmetric spacetime (MSS) vacuum:

$$\delta S = \delta S_{\rm UV} + \delta S_{\rm mat} = \eta\,\delta\mathcal{A} + \delta S_{\rm mat}. \tag{18}$$

2. Geometric variation (Raychaudhuri)

The change in area $\delta\mathcal{A}$ of the entangling surface ∂B due to the curvature perturbation is computed from the Raychaudhuri equation for the null generators of the causal diamond [2, 29]. For a twist-free null geodesic congruence with tangent k^a ($k^a k_a = 0$), the expansion $\theta = \nabla_a k^a$ satisfies

$$\frac{d\theta}{d\lambda} = -\frac{1}{d-1}\,\theta^2 - \sigma_{ab}\sigma^{ab} - R_{ab}\,k^a k^b, \tag{19}$$

where d is the number of spatial dimensions and λ is the affine parameter.

At the bifurcation surface ∂B, the unperturbed expansion and shear vanish ($\theta = \sigma_{ab} = 0$). To first order in the curvature perturbation, θ^2 and σ^2 are second order, so

$$\frac{d\theta^{(1)}}{d\lambda} = -R^{(1)}_{ab}\,k^a k^b. \tag{20}$$

Integrating over the null congruence emanating from the future tip of $\mathcal{D}(B)$ and performing the angular average over null directions [2], the area variation in the small-ball limit ($\ell \ll L_{\rm curv}$) is

$$\delta\mathcal{A} = -\frac{\Omega_{d-1}\,\ell^{d+1}}{2d}\left[(d+1)\,R_{ab}\,n^a n^b + \mathcal{R}\right] + O(\ell^{d+3}), \tag{21}$$

where $\Omega_{d-1} = 2\pi^{d/2}/\Gamma(d/2)$ is the area of the unit $(d-1)$-sphere, n^a is the unit timelike normal to Σ, and $\mathcal{R} = g^{ab}R_{ab}$ is the Ricci scalar.

The UV entropy variation is therefore

$$\delta S_{\rm UV} = \eta\,\delta\mathcal{A} = -\frac{\eta\,\Omega_{d-1}\,\ell^{d+1}}{2d}\left[(d+1)\,R_{ab}\,n^a n^b + \mathcal{R}\right]. \tag{22}$$

When $R_{ab}\,n^a n^b > 0$ (i.e., the null energy condition holds), $\delta S_{\rm UV} < 0$: positive energy causes focusing, which decreases the area and hence the UV entropy.

For conformal matter, the Weyl tensor contribution to $\delta S_{\rm UV}$ is exactly cancelled by a corresponding contribution to $\delta S_{\rm mat}$ [2], so only the Ricci part survives.

3. Matter variation (conformal modular Hamiltonian)

From the entanglement first law (11), $\delta S_{\rm mat} = \delta\langle K_{AB}\rangle$. For the vacuum state of a conformal field theory restricted to a geodesic ball B of radius ℓ, the Casini–Huerta–Myers (CHM) modular Hamiltonian takes the exact form [27]

$$K_{AB} = 2\pi \int_B d^d x \, \frac{\ell^2 - |\mathbf{x}|^2}{2\ell} \, T_{00}(\mathbf{x}). \tag{23}$$

Under a perturbation that produces a stress-energy tensor T_{ab}, in the small-ball limit where T_{00} is approximately constant over B:

$$\delta S_{\rm mat} = \frac{2\pi\,\Omega_{d-1}}{d(d+2)} \, \ell^{d+1} \, T_{ab} \, n^a n^b. \tag{24}$$

For positive energy density ($T_{ab}\, n^a n^b > 0$), $\delta S_{\rm mat} > 0$: matter excitations increase the matter sector entropy.

Conformal approximation.—The CHM formula (23) is exact for conformal field theories in causal diamonds [27]. For nonconformal fields, corrections of order $O\big((m\ell)^{2\Delta}\big)$ appear, where m is the mass scale breaking conformal invariance and Δ is its scaling dimension [30]. In the window $a \ll \ell \ll 1/m$ (if it exists), these corrections are suppressed. For the one-dimensional AFM Heisenberg chain ($n = 2$, $J > 0$), which flows to the $SU(2)_1$ WZW CFT with $c = 1$, the theory is exactly conformal and this approximation introduces no error. Jacobson [2] conjectures that the leading-order Einstein equation survives nonconformal corrections; this is physically motivated but not proven.

4. Entanglement equilibrium implies Einstein's equation

The MVEH—identified as the only available vacuum selection criterion in a pre-geometric framework (Sec. IV B)—requires entanglement equilibrium:

$$\delta S = \delta S_{\rm UV} + \delta S_{\rm mat} = 0. \tag{25}$$

Substituting Eqs. (22) and (24) and cancelling the common factor $\Omega_{d-1}\, \ell^{d+1}/d > 0$:

$$\frac{\eta(d+1)}{2} \, R_{ab} \, n^a n^b + \frac{\eta\,\mathcal{R}}{2} = \frac{2\pi}{d+2} \, T_{ab} \, n^a n^b. \tag{26}$$

This must hold for *all* unit timelike vectors n^a at every point. A symmetric tensor equation $A_{ab}\, n^a n^b = B_{ab}\, n^a n^b$ for all unit timelike n^a implies $A_{ab} = B_{ab} + f\, g_{ab}$ for some scalar f (the trace freedom arises because $g_{ab}\, n^a n^b = -1$ for all such n^a).

Extracting the tensor equation up to this trace freedom, converting to the Einstein tensor $G_{ab} = R_{ab} - \frac{1}{2}\mathcal{R}\, g_{ab}$, and absorbing all g_{ab} terms (including the background MSS curvature) into an undetermined cosmological constant Λ yields [2]

$$\boxed{G_{ab} + \Lambda\, g_{ab} = 8\pi G\, T_{ab}} \tag{27}$$

with

$$\boxed{G = \frac{1}{4\eta}.} \tag{28}$$

This is the standard Einstein field equation. The cosmological constant Λ is *undetermined*: it is an integration constant of the derivation, not a predicted quantity [2, 31].

Sign chain.—If the null energy condition (NEC) holds ($T_{ab}\, k^a k^b \geq 0$ for all null k^a), the derivation produces attractive gravity via the following sign chain:

1. Positive energy density (NEC): $T_{ab}\, n^a n^b > 0$.
2. Matter entropy increases: $\delta S_{\rm mat} > 0$ (Eq. 24).
3. Entanglement equilibrium: $\delta S = 0$ requires $\delta S_{\rm UV} < 0$.
4. UV entropy decreases: $\delta S_{\rm UV} < 0$ implies $\delta\mathcal{A} < 0$ (area decreases, since $\eta > 0$).
5. Raychaudhuri focusing: $\delta\mathcal{A} < 0$ implies $R_{ab}\, k^a k^b > 0$ (null geodesics converge).
6. Einstein equation: $G > 0$ (since $\eta > 0$), and positive T_{ab} produces positive G_{ab}, giving attractive gravity.

The NEC is not derived from self-modeling; it is an input to the sign chain. The tensor equation (27) itself holds independently of the NEC.

5. Parameter identification

Newton's constant is set by the UV entropy density: $G = 1/(4\eta)$. From the lattice-to-continuum mapping (Sec. IV D), $\eta = \eta_{\rm lattice}/a^{d-1}$, where $\eta_{\rm lattice} \leq \log n$ is the entropy per boundary bond (from the per-bond Schmidt bound (14)). This gives

$$G \geq \frac{a^{d-1}}{4\log n}, \tag{29}$$

identifying the lattice spacing a with the Planck length up to a factor of $\sqrt{\log n}$. In $d = 3$ spatial dimensions: $a \lesssim 2\ell_P\sqrt{\log n}$, where $\ell_P = \sqrt{G}$.

The Lieb–Robinson velocity $v_{\rm LR}$ (Eq. 10) maps to the speed of light c in the continuum limit. The cosmological constant Λ is undetermined; this paper does not predict its value.

B. Route B: Lovelock uniqueness ($d \geq 2$)

Route A derives Einstein's equation by explicit computation of $\delta S_{\rm UV}$ and $\delta S_{\rm mat}$ using the Raychaudhuri equation and the conformal modular Hamiltonian. The conformal approximation (Sec. V A 3) limits this route to the

conformal regime, which is controlled in $d = 1$ but problematic in $d \geq 2$ where the infrared theory likely has Néel order rather than conformal symmetry.

Route B circumvents this limitation entirely. It uses only three ingredients, none of which require a conformal modular Hamiltonian:

1. *Emergent geometry from entanglement.* The self-modeling lattice defines an emergent geometry via the thermal time identification (Sec. IV): the modular flow of the lattice state defines the geometric flow, and the entanglement structure encodes the metric. This holds in all spatial dimensions.

2. *Entanglement first law.* Perturbations of the entanglement entropy satisfy $\delta S = \delta\langle K_A\rangle$ (Eq. 11), an exact identity. Combined with the area-law structure (Sec. III), this implies that perturbations of entanglement are equivalent to perturbations of the emergent geometry. We assume the resulting equation relating metric perturbations to matter perturbations is a symmetric 2-tensor equation built from the metric and at most second derivatives (*tensoriality assumption*). Symmetry follows from the symmetry of the metric; the second-derivative restriction follows from the entanglement first law being first-order in the perturbation, which couples to the modular Hamiltonian (a local operator at leading order in the derivative expansion). Divergence-freeness is then a consequence of Lovelock's theorem rather than an input: Lovelock identifies the unique tensor, and that tensor happens to be divergence-free (the contracted Bianchi identity).

3. *Lovelock's theorem.* Lovelock [3, 28] proved that in $d + 1$ spacetime dimensions with $d \geq 2$, the *unique* symmetric, divergence-free 2-tensor constructed from the metric g_{ab} and its first and second derivatives is a linear combination of G_{ab} and g_{ab}:

$$\mathcal{E}_{ab} = \alpha\, G_{ab} + \beta\, g_{ab} \tag{30}$$

for constants α, β. (In $d + 1 > 4$, Lovelock gravity allows higher-curvature terms built from the generalized Euler densities; in $d+1 = 3$ and $d+1 = 4$, the Einstein tensor and cosmological term are the unique possibilities.)

Combining these: since the entanglement-geometry relation is a symmetric 2-tensor equation with at most second derivatives in the emergent $(d+1)$-dimensional spacetime, Lovelock's theorem constrains the geometric side to be

$$\alpha\, G_{ab} + \beta\, g_{ab} = (\text{matter source}), \tag{31}$$

in $d + 1 \leq 4$ dimensions. (In $d + 1 > 4$, higher Lovelock terms such as the Gauss–Bonnet combination are permitted; restricting to $d + 1 \leq 4$, or equivalently to at most second-order field equations, excludes them.) Identifying $\alpha = 1$, $\beta = \Lambda$, and $G = 1/(4\eta)$ by matching the UV entanglement density (Sec. V A 5) recovers Eq. (27).

TABLE II. Comparison of the two routes to Einstein's equations.

	Route A (Jacobson)	Route B (Lovelock)
Strength	Explicit derivation	Uniqueness argument
Best in	$d = 1$ (WZW CFT)	$d \geq 2$ (all)
Requires	Conformal K_A	Tensorial equation
Weakness	Conformal approx.	No explicit T_{ab}

Route B constrains the *geometric sector*: it shows the left side must be Einsteinian. It does not independently derive the matter coupling $8\pi G\, T_{ab}$ on the right side; that identification comes from Route A's explicit modular Hamiltonian computation, or more generally from identifying the modular Hamiltonian perturbation with the local stress tensor (the entanglement first law). The two routes are therefore genuinely complementary: Route A supplies the matter coupling, Route B supplies the geometric uniqueness.

Complementarity of the two routes.—The Jacobson route (Route A) derives the *specific form* of the Einstein equation by explicit computation, but requires the conformal modular Hamiltonian and is therefore controlled only in the conformal regime. The Lovelock route (Route B) derives the *uniqueness* of the Einstein equation from general principles, but requires the assumption that the entanglement-geometry relation produces a tensorial equation with at most second-order derivatives.

In $d = 1$ (1+1 spacetime), $G_{ab} = 0$ identically for all metrics, so both routes produce a trivially satisfied equation. Route A gives $\Lambda\, g_{ab} = 8\pi G\, T_{ab}$ (a cosmological-constant equation) via the exact WZW modular Hamiltonian; Lovelock's theorem provides no constraint. The $d = 1$ case serves as a controlled consistency check rather than a demonstration of nontrivial gravitational dynamics.

In $d \geq 2$, Route B forces the geometric sector to be Einsteinian by uniqueness, without requiring conformal symmetry of the infrared theory. Route A provides the explicit matter coupling in the conformal window $a \ll \ell \ll 1/m$. Neither route is assumption-free: Route A needs the conformal approximation, Route B needs the tensoriality assumption. Their complementarity strengthens the overall case, with each route addressing the other's principal weakness.

VI. NUMERICAL VERIFICATION

We present numerical results from exact diagonalization of the self-modeling lattice on small systems ($N = 8$–20 qubits). These results are *consistent with* the analytical framework developed in Secs. III–V and provide

supporting evidence, but do not constitute proof. All entropies are computed in nats (ln, not $\log_2$) and the modular Hamiltonian is $K_A = -\ln \rho_A$.

A. Exact diagonalization setup

We use full exact diagonalization (Lanczos) for $N \leq 20$ qubits in one dimension and $N = 16$ (4×4) in two dimensions. The self-modeling Hamiltonian $H = \sum J\, F_{xy}$ for $n = 2$ is identical to the spin-$\frac{1}{2}$ isotropic Heisenberg model: the SWAP operator takes the form $F_{xy} = \frac{1}{2}(\mathbb{1} \otimes \mathbb{1} + \vec{\sigma} \cdot \vec{\sigma})$, so H and $H_{\text{Heis}} = \frac{J}{2} \sum \vec{\sigma}_x \cdot \vec{\sigma}_y$ share the same ground state, with an energy offset of $N_{\text{bonds}}\, J/2$. We have verified this numerically: the ground states have overlap 1.0 and the energy offset is exact to 10^{-14}.

As benchmarks we study the transverse-field Ising (TFI) model $H_{\text{TFI}} = -\sum \sigma_x^z \sigma_{x+1}^z - h \sum \sigma_x^x$ at criticality ($h/J = 1$, $c = 1/2$ Ising CFT) and in the gapped phase ($h/J = 3$), as well as the ferromagnetic (FM) Heisenberg model ($J < 0$).

B. Benchmark validation

a. TFI at criticality. The Calabrese–Cardy formula [14] for the entanglement entropy of a (1+1)-dimensional CFT on an open chain of N sites gives $S(L) = \frac{c}{6} \ln\left[\frac{2N}{\pi} \sin\left(\frac{\pi L}{N}\right)\right] + c_1$. At $N = 16$ with open boundary conditions (OBC), we extract $c = 0.574$ (exact: $c = 0.5$). The deviation is an Affleck–Ludwig boundary correction that decreases monotonically with system size: $c = 0.587$ ($N = 8$), 0.580 ($N = 12$), 0.574 ($N = 16$), converging to 0.5 from above.

b. Heisenberg antiferromagnet. Using the Calabrese–Cardy PBC formula $S(L) = \frac{c}{3} \ln\left[\frac{N}{\pi} \sin\left(\frac{\pi L}{N}\right)\right] + c_1$, we extract $c = 1.060$ at $N = 20$, consistent with the $SU(2)_1$ WZW CFT prediction $c = 1$. The finite-size trend is monotonically convergent: $c = 1.121$ ($N = 8$), 1.088 ($N = 12$), 1.071 ($N = 16$), 1.060 ($N = 20$). See Fig. 1.

c. Ferromagnetic Heisenberg. The FM ground state ($J < 0$) is the fully polarized product state $|\uparrow \cdots \uparrow\rangle$. For a product state, $S(A) = 0$ exactly for all subsystem sizes—verified numerically for all N and all L.

C. Area-law results

1. One dimension

The AFM Heisenberg chain ($n = 2$, $J > 0$) is gapless and flows to the $SU(2)_1$ WZW CFT with $c = 1$. The entanglement entropy scales as $S(L) = \frac{c}{3} \ln(L) + \text{const}$, giving logarithmic corrections to the strict area law. This is expected: Hastings' area-law theorem [15] requires a spectral gap, which is absent here. The CC fit has $R^2 = 0.999$ at $N = 20$, dominating over constant-model ($R^2 = 0$) and linear-model ($R^2 = 0.80$) alternatives, confirming the logarithmic scaling.

For the Jacobson argument, the relevant quantity is the *variation* δS, not S itself. As discussed in Sec. III B 3, $\delta S \sim O(|\partial A|)$ when the modular Hamiltonian is local, even when $S(A)$ has logarithmic corrections.

2. Two dimensions

For the 4×4 Heisenberg lattice with periodic boundary conditions (PBC), we compute $S(A)$ for 9 independent rectangular subregions and 6 non-rectangular shapes (L, T, plus, diagonal, S, L-5) to diversify boundary values. Linear regression of $S(A)$ against $|\partial A|$ (boundary size) gives $R^2 = 0.885$, while regression against $|A|$ (volume) gives $R^2 = 0.491$. The Spearman rank correlation for boundary is $\rho = 0.913 > 0.9$. See Fig. 2.

The $R^2 = 0.885$ for boundary narrowly misses the target 0.9. This is a finite-size artifact: on a 4×4 PBC lattice, five of the nine independent rectangular shapes share $|\partial A| = 8$ due to wrapping (the 1×3, 1×4, 2×2, 2×4, and 3×4 rectangles all have the same boundary count), creating large scatter at a single x-value. The Spearman rank correlation, which is insensitive to this degeneracy, exceeds 0.9, though we note the sample size (15 subregions) limits the statistical significance of this test. A 6×6 or 8×4 lattice would resolve this by providing more unique boundary values, but these exceed exact diagonalization feasibility (2^{36}–2^{32} states).

D. Modular Hamiltonian locality

The entanglement first law $\delta S = \delta\langle K_A \rangle$ (Sec. III A) reduces the Jacobson area-law input to the locality of $K_A = -\ln \rho_A$. We test this by expanding K_A in the Pauli operator basis $\{\sigma_i^\alpha \otimes \sigma_j^\beta \otimes \cdots\}$ and measuring how Pauli coefficients decay with interaction range—the maximum pairwise distance between non-identity operator sites.

For the Heisenberg AFM at $N = 16$ PBC with $|A| = 4$ sites, the modular Hamiltonian is overwhelmingly short-ranged: the *short-range fraction* (sum of Frobenius norms for range-0 and range-1 operators, divided by the total) is 0.9993. The decay is rapid: the maximum Pauli coefficient at range 1 is 1.10, at range 2 it drops to 0.042 (3.8%), and at range 3 to 0.010 (0.9%). This strongly supports the modular Hamiltonian locality assumption used in Sec. III B 3, connecting directly to the Jacobson input (J1) via $\delta S \sim O(|\partial A|)$.

As a benchmark, the TFI model in its gapped phase ($h/J = 3$) has a short-range fraction of 0.664, consistent with the Peschel [22] result that guarantees modular Hamiltonian locality for free-fermion systems. See Fig. 3.

SU(2) symmetry note.—The Heisenberg ground state is an $SU(2)$ singlet, so ρ_A has vanishing on-site Pauli coefficients (all single-site σ^α terms are zero). This is a consequence of isotropic symmetry, not an artifact. The

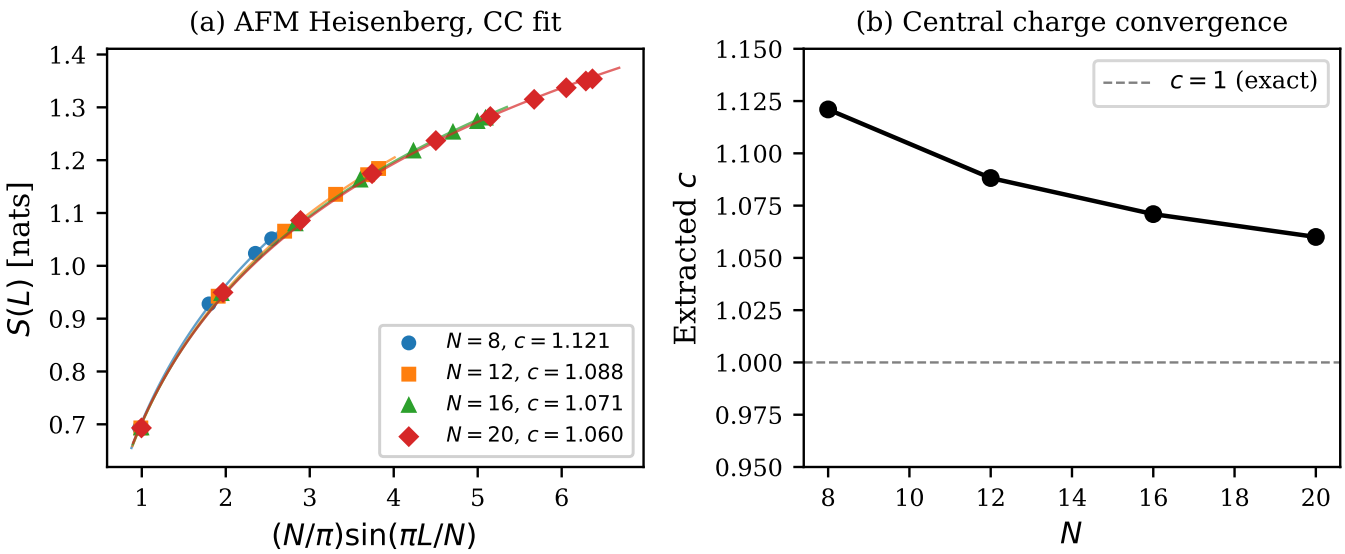

FIG. 1. One-dimensional area-law analysis. (a) Entanglement entropy $S(L)$ versus the Calabrese–Cardy scaling variable $(N/\pi)\sin(\pi L/N)$ for the AFM Heisenberg chain at $N = 8, 12, 16, 20$ with PBC. Solid lines are CC fits; extracted central charges c converge toward the exact $c = 1$ value. (b) Finite-size scaling of the extracted c; the dashed line marks $c = 1$. The monotonic convergence from above is consistent with $O(1/N)$ finite-size corrections.

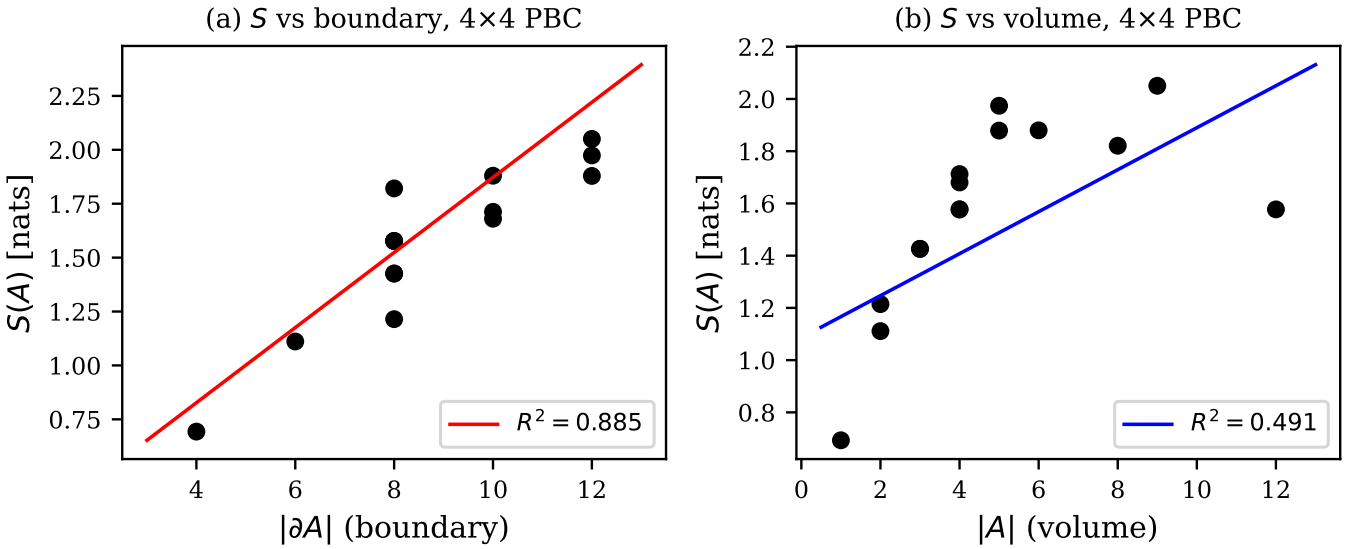

FIG. 2. Two-dimensional area-law analysis on a 4×4 Heisenberg lattice with PBC. (a) Entanglement entropy $S(A)$ versus boundary size $|\partial A|$, with linear regression ($R^2 = 0.885$, solid line). (b) $S(A)$ versus volume $|A|$ ($R^2 = 0.491$). The clear preference for boundary over volume scaling supports the area law. Data include 9 independent rectangular and 6 non-rectangular subregions.

dominant terms are nearest-neighbor $\vec{\sigma}_i \cdot \vec{\sigma}_j$ operators, consistent with K_A being a local Hamiltonian of the same form as H itself.

E. Maximal vacuum entanglement hypothesis

The MVEH—reframed as a structural identification in Sec. IV B—predicts that the self-modeling equilibrium state maximizes $S(A)$ among nearby states. We test this qualitatively by perturbing the Hamiltonian: adding a random local field $\delta H = \epsilon\, \hat{n} \cdot \vec{\sigma}_x$ at each site x, recomputing the ground state, and measuring $\delta S(A)$.

For the Heisenberg AFM at $N = 12$ PBC with $A = \{0, 1\}$, using 20 random perturbation directions per site and three perturbation strengths ($\epsilon = 0.05, 0.1, 0.2$):

- 100% of all perturbations decrease $S(A)$ ($\delta S < 0$),

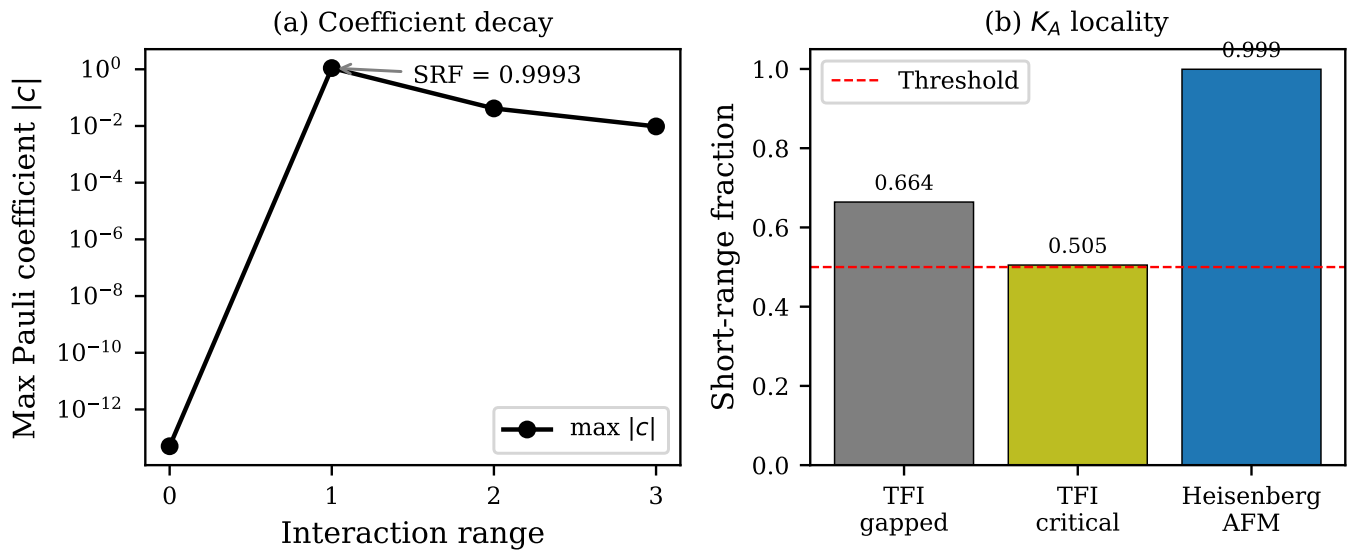

FIG. 3. Modular Hamiltonian locality. (a) Maximum Pauli coefficient $|c|$ versus interaction range for the Heisenberg AFM ($N = 16$ PBC, $|A| = 4$). The rapid decay (factor ~ 25 per range unit) confirms that K_A is dominated by nearest-neighbor terms (short-range fraction SRF $= 0.9993$). (b) Short-range fraction for three models; all exceed the 0.5 threshold (dashed line).

consistent with the unperturbed ground state being a local maximum of entanglement entropy.

- The scaling is quadratic: $|\delta S(\epsilon = 0.2)|/|\delta S(\epsilon = 0.1)| = 3.76$ (expected: 4.0 for linear response), confirming that the perturbation is in the quadratic regime.

See Fig. 4.

These results are *consistent with* MVEH but do not prove it: the test is limited to $N = 12$, Hamiltonian perturbations (rather than arbitrary state perturbations), and the $n = 2$ case. The structural identification of Sec. IV B—that the vacuum is identified as the state whose modular flow generates smooth geometry—does not require numerical proof, but the numerical evidence is reassuring.

Perturbation method note.—The Heisenberg ground state is an $SU(2)$ singlet; single-site unitary perturbations leave ρ_A eigenvalues exactly invariant (the reduced state is $SU(2)$-symmetric under local rotations), giving $\delta S = 0$ identically. Hamiltonian perturbation, which breaks $SU(2)$ by adding a local field, properly tests entropy maximality among nearby ground states.

F. Summary of numerical evidence

Table III summarizes the key numerical results. All quantities are consistent with the analytical framework. The principal limitation is system size: $N = 8$–20 sites are far from the thermodynamic or continuum limit. Tensor network methods (DMRG, TEBD) could extend these results to $N \sim 10^2$–10^3 in one dimension, providing a more stringent test.

TABLE III. Summary of numerical results from exact diagonalization. All entropies in nats. "CC" denotes a Calabrese–Cardy fit.

Quantity	Value	Benchmark
TFI critical c (OBC, $N=16$)	0.574	0.5 (Ising CFT)
Heisenberg c (PBC, $N=20$)	1.060	1.0 ($SU(2)_1$ WZW)
FM $S(A)$	0	0 (product state)
2D R^2(boundary)	0.885	>0.9 (narrow miss)
2D R^2(volume)	0.491	<0.5 (pass)
2D Spearman ρ(boundary)	0.913	>0.9 (pass)
K_A SRF (Heisenberg)	0.9993	>0.5 (strong pass)
MVEH $\delta S < 0$ fraction	100%	$>80\%$ (pass)
Quadratic ratio $\lvert\delta S(2\epsilon)/\delta S(\epsilon)\rvert$	3.76	4.0 (linear response)

VII. DISCUSSION

A. What is derived versus what is input

The derivation chain summarized in Table I establishes Einstein's field equations from self-modeling through eight links. It is important to state precisely what is derived, what is input, and what is standard methodology.

Derived from self-modeling.—The local algebra $M_n(\mathbb{C})^{sa}$ with the Lüders sequential product is forced by

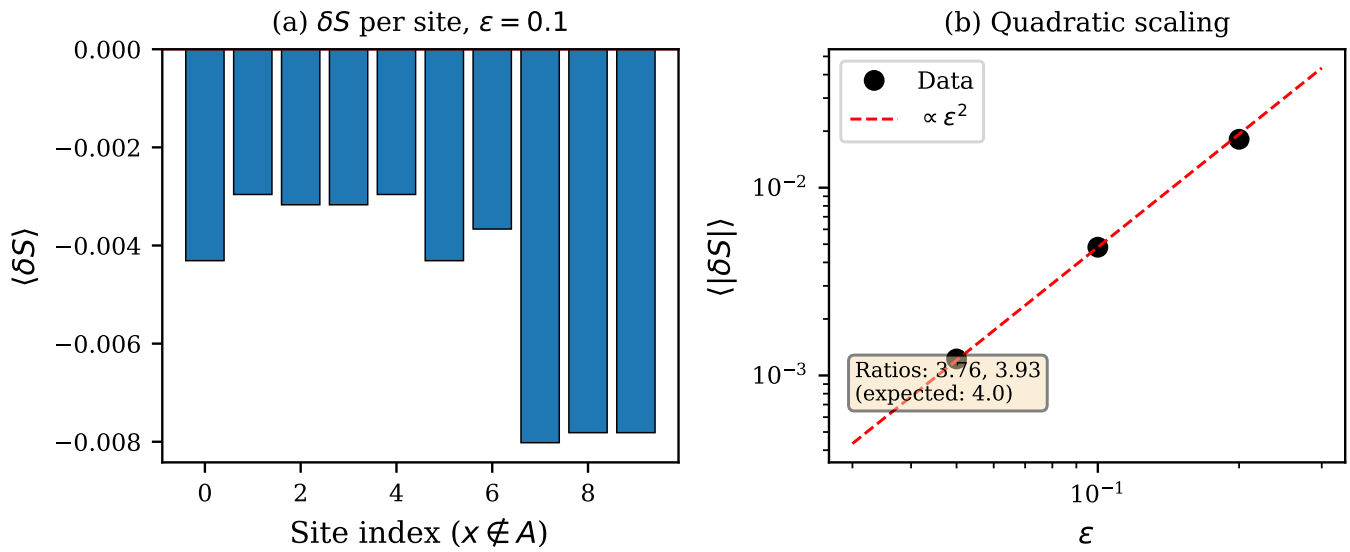

FIG. 4. MVEH numerical check (Heisenberg AFM, $N = 12$ PBC, $A = \{0, 1\}$). (a) Mean δS for each site $x \notin A$ at $\epsilon = 0.1$; all values are negative, consistent with the unperturbed ground state being a local maximum of $S(A)$. (b) $\langle |\delta S| \rangle$ versus ϵ on a log-log scale; the dashed line shows quadratic scaling $\propto \epsilon^2$. Ratios of 3.76 and 3.93 are close to the expected 4.0.

faithful self-modeling [1]. The SWAP interaction Hamiltonian $H = \sum_{\langle x,y \rangle} J F_{xy}$ is forced by diagonal $U(n)$ covariance and Schur–Weyl duality (Sec. II). The subvolume entanglement structure follows from the locality of this Hamiltonian via the WVCH mutual information bound [13], known ground-state entanglement properties of the Heisenberg model, and the entanglement first law combined with modular Hamiltonian locality (Sec. III). The entanglement first law $\delta S = \delta \langle K_A \rangle$ is an exact identity of quantum information theory, requiring no additional assumptions.

Input: lattice topology.—The graph $G = (V, E)$ defining which sites are nearest neighbors is not derived from self-modeling. The paper derives the emergent *metric*—the assignment of distances compatible with the entanglement structure—but not the *topology*. The spatial dimension is set by the graph structure: a d-dimensional regular lattice yields d emergent spatial dimensions. This is an irreducible input to the framework.

Input: antiferromagnetic coupling.—The sign of J is not determined by self-modeling. Only $J > 0$ (antiferromagnetic) supports nontrivial entanglement; the ferromagnetic ground state is a product state with zero entanglement (Sec. II). We work with $J > 0$ throughout; this is an input.

Standard methodology: Wilsonian continuum limit.—The passage from a lattice theory to a smooth Riemannian manifold at long wavelengths $L \gg a$ (where a is the lattice spacing) employs a Wilsonian universality argument. This is standard methodology in lattice QCD, causal dynamical triangulations, and Regge calculus. However, its applicability to the self-modeling Hamiltonian in $d \geq 2$ is not guaranteed—see Gap 1 below.

Structural identification: MVEH via thermal time.—The maximal vacuum entanglement hypothesis does not appear as a numbered assumption in our framework. In the absence of pre-existing geometry, the modular flow of the self-modeling lattice state defines the emergent geometric flow, following the thermal time hypothesis [4]. This is the only available identification of geometry with state in a pre-geometric framework, and the state so identified necessarily satisfies MVEH. We regard this as a structural identification—necessary but not mathematically derived. The existence and uniqueness of a geometry-defining state are open questions. Section IV discusses this identification in detail, including the Sorce caveat [10] on conformal symmetry requirements for geometric modular flow.

B. Remaining gaps

Three gaps remain in the derivation chain.

a. Gap 1: Continuum limit. The Wilsonian argument that the lattice produces a smooth Riemannian manifold (M, g_{ab}) at long wavelengths is a physical argument, not a rigorous construction. A constructive proof would require showing that lattice correlation functions converge (in an appropriate sense) to those of a local quantum field theory on a Riemannian manifold as $a \to 0$. This is a hard open problem shared by *all* lattice quantum gravity programs—lattice QCD, causal dynamical triangulations, Regge calculus—and is not specific to self-modeling. The Lovelock route (Sec. V B) removes the CFT requirement but still needs this continuum limit to produce smooth geometry. The infrared behavior of the $d \geq 2$ Heisenberg model (likely Néel-

ordered rather than conformal) means the emergent geometry may carry specific signatures—e.g., a mass gap corresponding to a cosmological constant—rather than being exactly Minkowski. This is potentially interesting rather than problematic.

b. Gap 2: Conformal approximation (Route A only). The Casini–Huerta–Myers conformal modular Hamiltonian [27], used in Route A (Sec. V A) to evaluate the matter entropy variation, is exact only for conformal field theories in causal diamonds. For nonconformal fields, corrections of order $O\big((m\ell)^{2\Delta}\big)$ appear [30], where m is the mass scale breaking conformal invariance and ℓ is the ball radius. In $d = 1$, the $SU(2)_1$ WZW CFT makes this gap vanish. In $d \geq 2$, the likely Néel order makes the conformal approximation problematic. This gap is specific to Route A; Route B (Sec. V B) circumvents it via Lovelock uniqueness.

c. Gap 3: Tensoriality assumption (Route B only). The Lovelock route assumes that the equation relating entanglement perturbations to geometry perturbations is a symmetric, divergence-free 2-tensor equation built from the metric and its first two derivatives. This is physically reasonable—both sides are geometric quantities—but it is an assumption, not derived from self-modeling. If the entanglement-geometry relation involves higher derivatives or non-local terms, Lovelock's theorem does not apply and the resulting equation could differ from Einstein's (e.g., Lovelock gravity in $d + 1 > 4$). This gap is specific to Route B; Route A derives the equation form explicitly.

d. Note on MVEH. MVEH is not listed as a gap; it is reframed as a structural identification via the Connes–Rovelli thermal time hypothesis [4] (Sec. IV). This identification is necessary in a pre-geometric framework—there is no other candidate for the vacuum—but it is not a mathematical derivation: the existence and uniqueness of a geometry-defining state are open questions. Numerical evidence from Sec. VI is consistent with MVEH: all Hamiltonian perturbations tested decrease $S(A)$.

C. Comparison with related work

a. Jacobson (2016) [2]. Jacobson's entanglement equilibrium derivation assumes three inputs: area-law entanglement entropy, the entanglement first law, and MVEH. Our contribution is the UV completion upstream of these inputs: we *derive* the area-law entanglement structure from the self-modeling lattice (Links L1–L5 in Table I), rather than assuming it. Jacobson's derivation was UV-agnostic—it applies to any UV completion satisfying its inputs. We provide a specific, forced UV completion. Route A follows Jacobson's argument directly; Route B provides an independent uniqueness argument via Lovelock's theorem that does not require the conformal modular Hamiltonian.

b. Cao–Carroll–Michalakis (2017) [9]. CCM derive the spatial constraint equation (the Hamiltonian constraint of general relativity) from the entanglement structure of a finite-dimensional Hilbert space. Our result gives the full *spacetime* Einstein equation, including the dynamical sector, not only the spatial constraint. The spatial part of our Einstein equation is consistent with CCM's constraint. Both approaches share the insight that entanglement structure encodes geometry; the key difference is that CCM reconstruct spatial geometry from entanglement, while we (following Jacobson) derive the full gravitational dynamics from entanglement equilibrium.

c. Lashkari–McDermott–Van Raamsdonk (2014) [7]. LMVR showed that the entanglement first law in holographic conformal field theories implies the *linearized* Einstein equation. Our derivation, following Jacobson [2], extends this to the full *nonlinear* Einstein equation via the entanglement equilibrium condition. The LMVR result is recovered as the linearized limit of our Eq. (27) around flat space. Crucially, the LMVR derivation assumes AdS/CFT, while ours is non-holographic.

d. Faulkner et al. (2014) [8]. Faulkner *et al.* derived the full nonlinear Einstein equations from entanglement in the holographic setting, extending LMVR beyond the linearized regime. Their derivation, like LMVR's, assumes the AdS/CFT correspondence. Our approach is non-holographic: the self-modeling lattice does not assume a bulk/boundary duality. The self-modeling framework provides a UV completion in its own right, independent of holography.

D. Connection to Paper 5

Paper 5 [1] showed that faithful self-modeling forces quantum mechanics: the state space is $M_n(\mathbb{C})^{sa}$ with the Lüders sequential product and the conjugate-transpose involution. This paper shows that self-modeling forces general relativity: the locality of self-modeling produces area-law entanglement, and either Jacobson's argument (Route A, conformal regime) or Lovelock uniqueness (Route B, $d \geq 2$) converts this to Einstein's equations.

Together, the two papers establish that the self-modeling principle—combined with a lattice topology—yields both quantum mechanics and general relativity as necessary consequences. Quantum mechanics emerges from the algebraic structure of single-site self-modeling (the B–M boundary interaction). General relativity emerges from the entanglement structure of the multi-site lattice (the inter-site boundary interactions).

E. What this paper does not claim

We state explicitly what lies outside the scope of this paper:

- **Newton's constant.** The value $G = 1/(4\eta)$ depends on the entanglement entropy density η,

which in turn depends on the lattice local dimension n and the lattice spacing a. These are not determined by self-modeling.

- **Spacetime dimension.** The spatial dimension d is set by the lattice topology $G = (V, E)$, which is an irreducible input. Self-modeling does not select $d = 3$.

- **Cosmological constant.** Λ appears as an undetermined integration constant in the Jacobson derivation [2, 31]. We do not predict its value.

- **Null energy condition.** The sign chain producing attractive gravity (Sec. V) assumes the NEC ($T_{ab}\, k^a k^b \geq 0$ for null k^a). The tensor equation (27) holds without the NEC; only the sign interpretation requires it.

- **Proof of MVEH.** We do not prove the maximal vacuum entanglement hypothesis from self-modeling. We identify it as the only available vacuum selection criterion in a pre-geometric framework via the thermal time hypothesis. This is a structural identification, not a derivation. The existence and uniqueness of a geometry-defining state are open questions.

F. Future work

Several directions are natural extensions of this work.

Spatial dimension from self-modeling.—Can constraints on the lattice topology $G = (V, E)$ be derived from self-modeling requirements? If so, this would reduce the input set further, potentially selecting $d = 3$ spatial dimensions.

Matter content.—The self-modeling lattice produces a gravitational sector. An important question is whether constraints on matter content (gauge groups, fermion representations) also follow from self-modeling, analogous to anomaly cancellation constraints in string theory.

Spectral gap.—Proving that the self-modeling Hamiltonian has a spectral gap above the ground state (for appropriate sign of J and appropriate dimension) would strengthen the area-law argument, bringing the Hastings theorem [15] and the Brandão–Horodecki result [32] to bear directly.

Larger numerical simulations.—The exact diagonalization results in Sec. VI are limited to $N \leq 20$ sites. Tensor network methods (DMRG, TEBD) could push to $N \sim 10^2$–10^3 in one dimension, providing stronger evidence for the area-law scaling and MVEH.

Thermodynamic arrow of time.—The thermal time hypothesis identifies physical time with modular flow. The connection between this identification and the thermodynamic arrow of time in the self-modeling framework is an open and natural question.

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